Regression Analysis of Energy Efficiency for Building Designs

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1 Introduction

As population keep growing, the amount of construction for more building that includes residential, business, and dedicated spaces increases accounting for substantial portion of energy consumption. Therefore, the early stages of on architectural decision for building geometry, glazing area, and height can determine heating and cooling loads before HVAC equipment is selected. The purpose of this regression analysis is to understand and predict how architectural designs variables can influence energy requirements combining statistical analysis as well as predictive machine learning models.

This analysis is conducted using **Energy Efficiency dataset from the UCI Machine Learning Repository**, which has 768 building samples with each sample containing two energy performance, one for heating which we are going to denote as $Y1$ and the other one for cooling $Y2$. The analysis combines Exploratory Data Analysis (EDA), classical statistical modeling (Linear Regression), and modern machine learning approaches (Lasso, Ridge, Random Forest). Then, deep analysis is made to understand which model performs best using metrics such as $\mathbf{R^2}$, Mean Absolute Error **(MAE)**, and Root Mean Squared Error **(RMSE)**. At the end we conclude our model comparison as well as suggestions based on the analysis.

2 Dataset Description

Each observation corresponds to a building configuration. The dependent (Y1 and Y2) independent variables and their meanings are shown in Table 1.

Table 1: Input and Output Variables

Variable	Description
Relative Compactness	Volume-to-surface ratio: higher values = more compact
Surface Area	Total exterior surface area (m^2)
Wall Area	Area of exterior walls (m^2)
Roof Area	Roof surface area (m^2)
Overall Height	Building height in meters
Orientation	Directional orientation (2: North, 3: East, 4: South, 5: West)
Glazing Area	Window area as percentage of floor area (0–40%)
Glazing Area Distribution	Window placement index (0–5)
Heating Load (Y1)	Annual heating energy requirement (kWh/m^2)
Cooling Load (Y2)	Annual cooling energy requirement (kWh/m^2)

All features are numerical except Orientation and Glazing Area Distribution, which are categorical. The dataset is complete and contains no missing values.

3 Exploratory Data Analysis

Exploratory Data Analysis revealed several important structural relationships:

3.1 Histograms

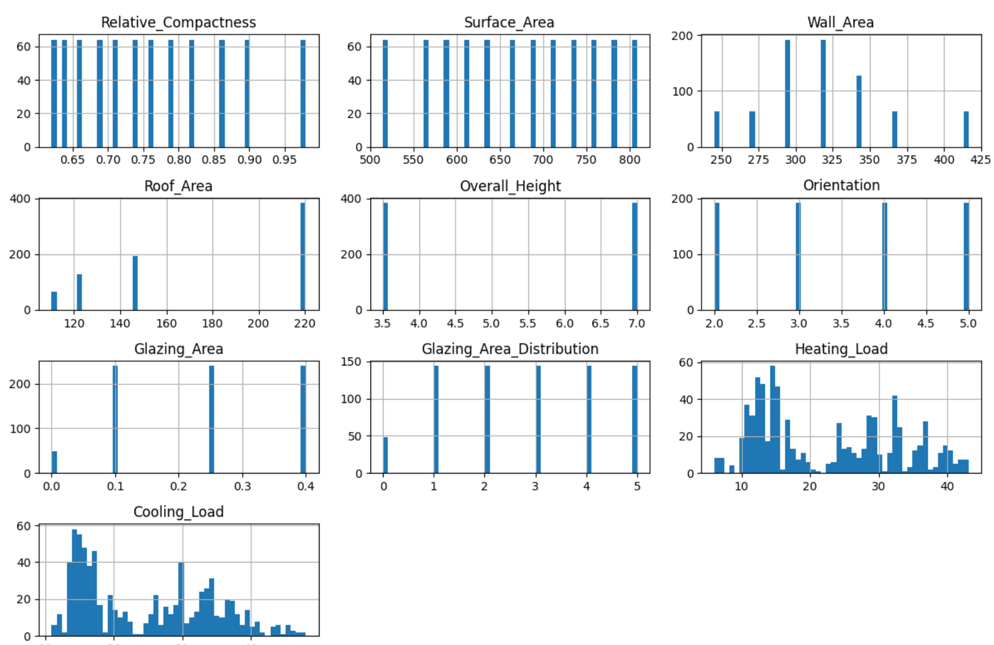

Figure 1: Histograms of Energy Efficiency Variables

When we take a look at our histograms specially our dependent variables we can see that they look unimodal but right-skewed which can mean that most buildings design perform within a moderate energy range while only a part/subset requires incredible large energy.

3.2 Correlation Structure

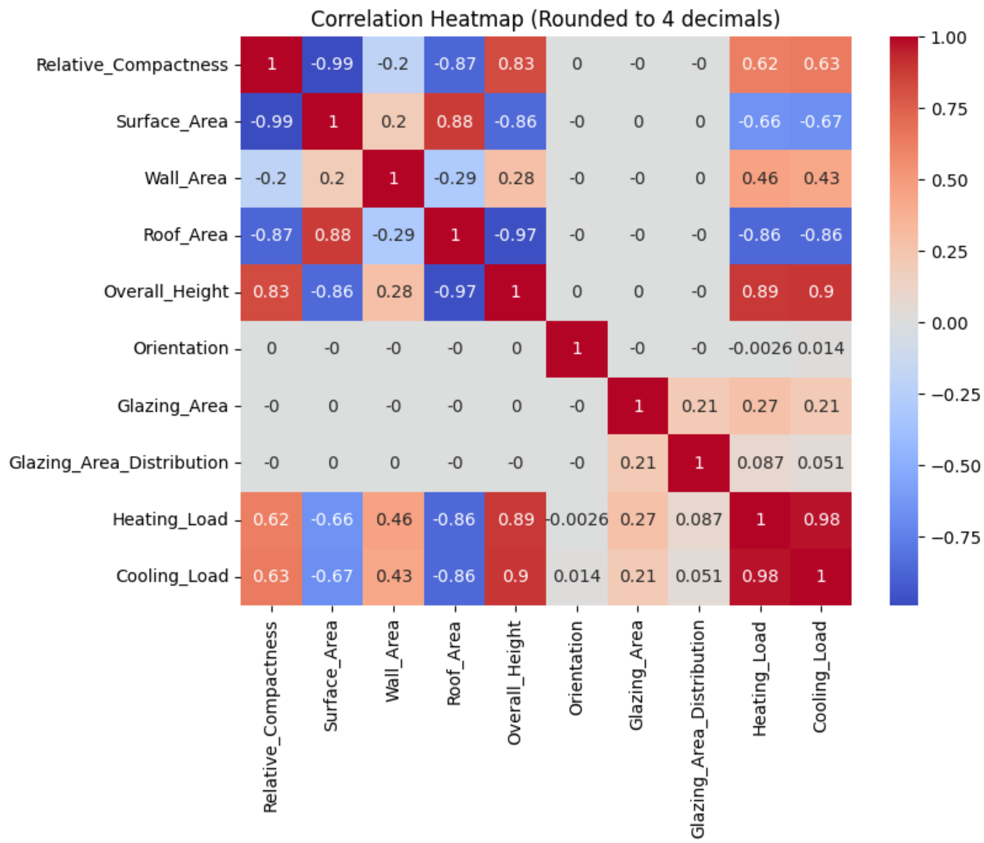

Figure 2: Correlation Matrix of Energy Efficiency Variables

The correlation matrix shows two dominant patterns:

1. **Relative Compactness** has a strong negative correlation with both heating and cooling loads. Buildings with higher compactness expose less surface area.

2. **Overall Height and Glazing Area** correlate positively with both Y1 and Y2. Increasing height or glazed surface increases the building envelope exposure, leading to higher loads.

The rest of the variables such as Orientation and Glazing Area Distribution exhibit near-zero correlations with the target variables, suggesting minimal influence on performance.

3.3 Scatter Plots

Scatter plots are divided into two sections for better interpretability, one section is about the geometry variables of our model which includes Relative Compactness, Wall Area Roof Area and Surface area, where the other sections has architectural features such as Overall Height, Orientation, Glazing Area and Distribution.
These sections are then compared to our dependent variable for Heating (Y1) and Cooling (Y2). After analyzing our scatter plots with in our independent variables with the dependent variables we found some interesting results that tell us more about the data.

3.3.1 Heating

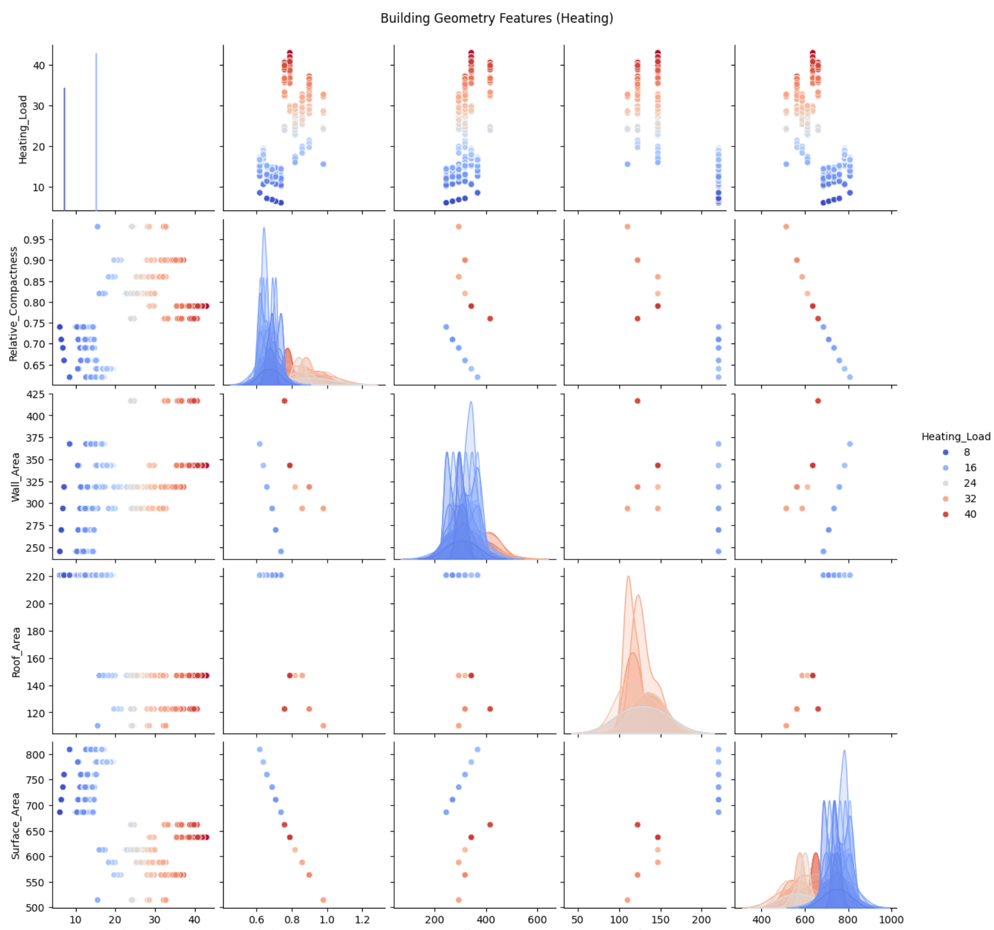

Figure 3: Scatter Plot for Geometry Features (Heating)

After analyzing Figure 3 we can see that surface ares has a strong correlation with other variables but surface area is also the total sum of the total area which can make our model redundant and dropping this variable will not impact our model.

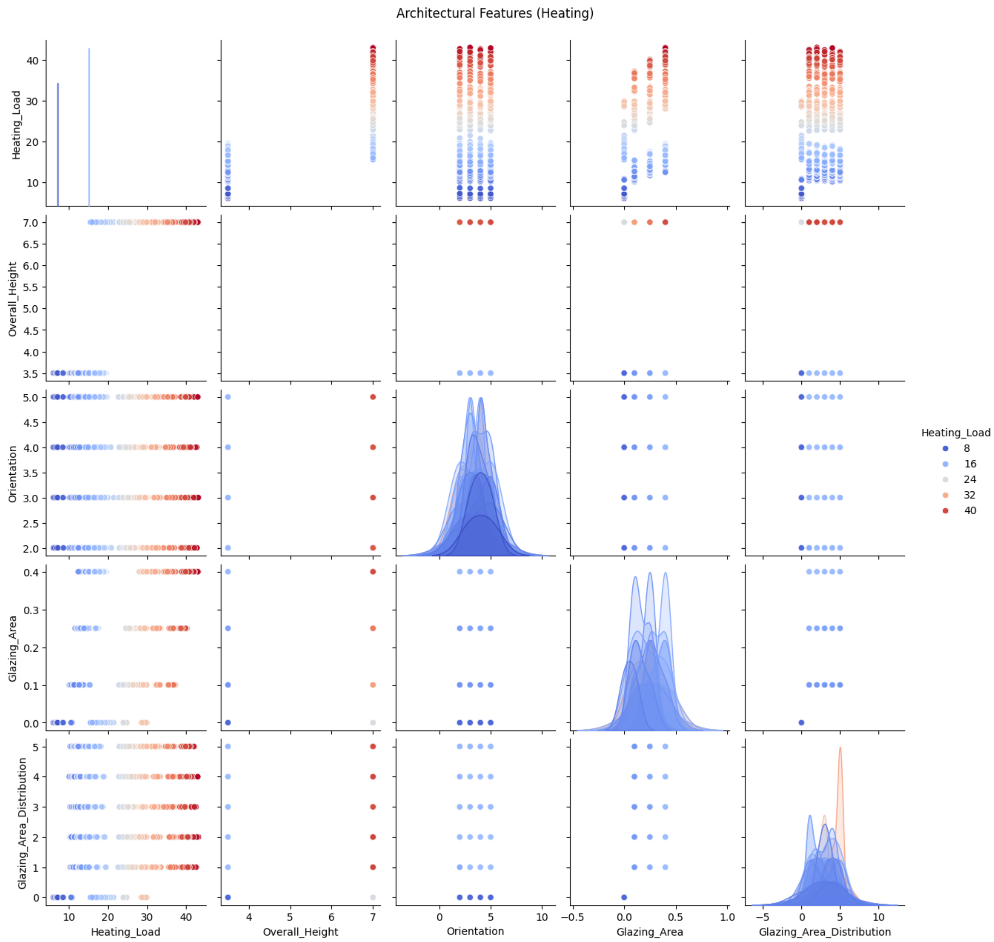

Figure 4: Scatter Plot for Architectural Features (Heating)

When we analyze Figure 4 for the Architectural Features, the independent variable Orientation has a weak or not meaningful correlation with heat, but further analysis is performed to decided how it impacts our model. Height has a moderate to positive correlation as well as the glazing area when compared to our target variable.

3.3.2 Cooling

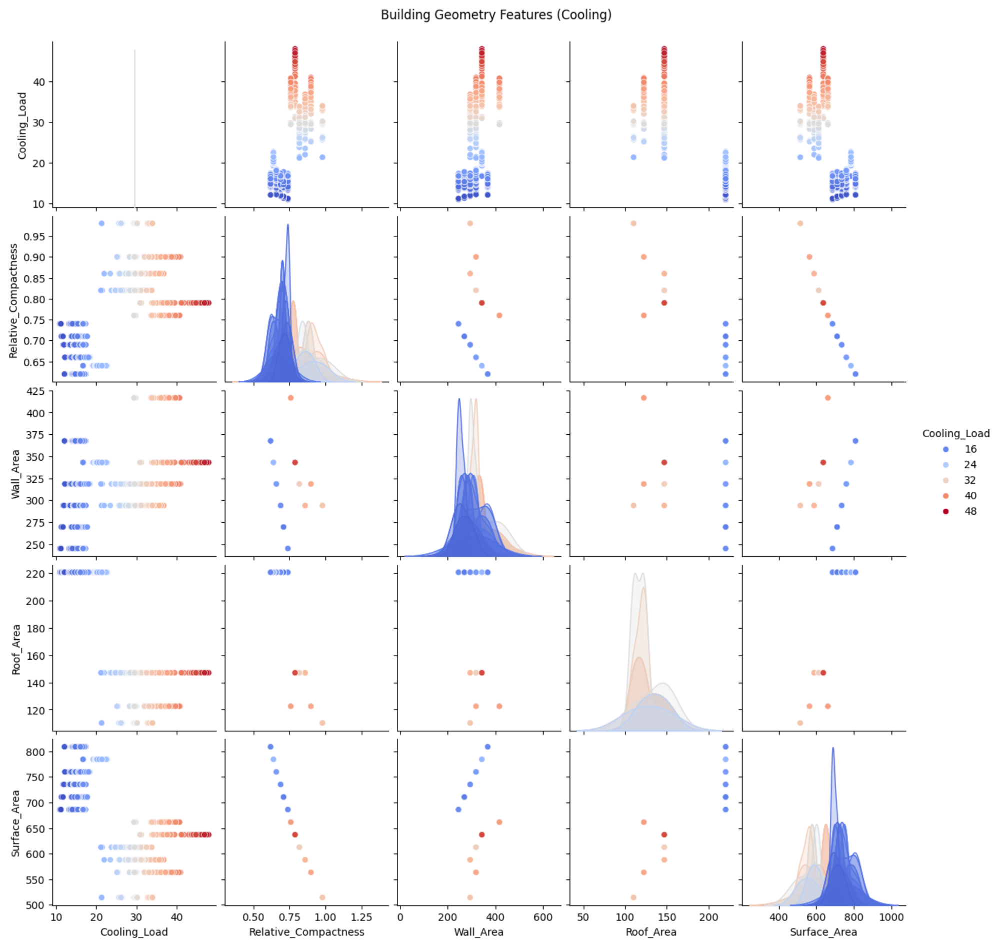

Figure 5: Scatter Plot for Geometry Features (Cooling)

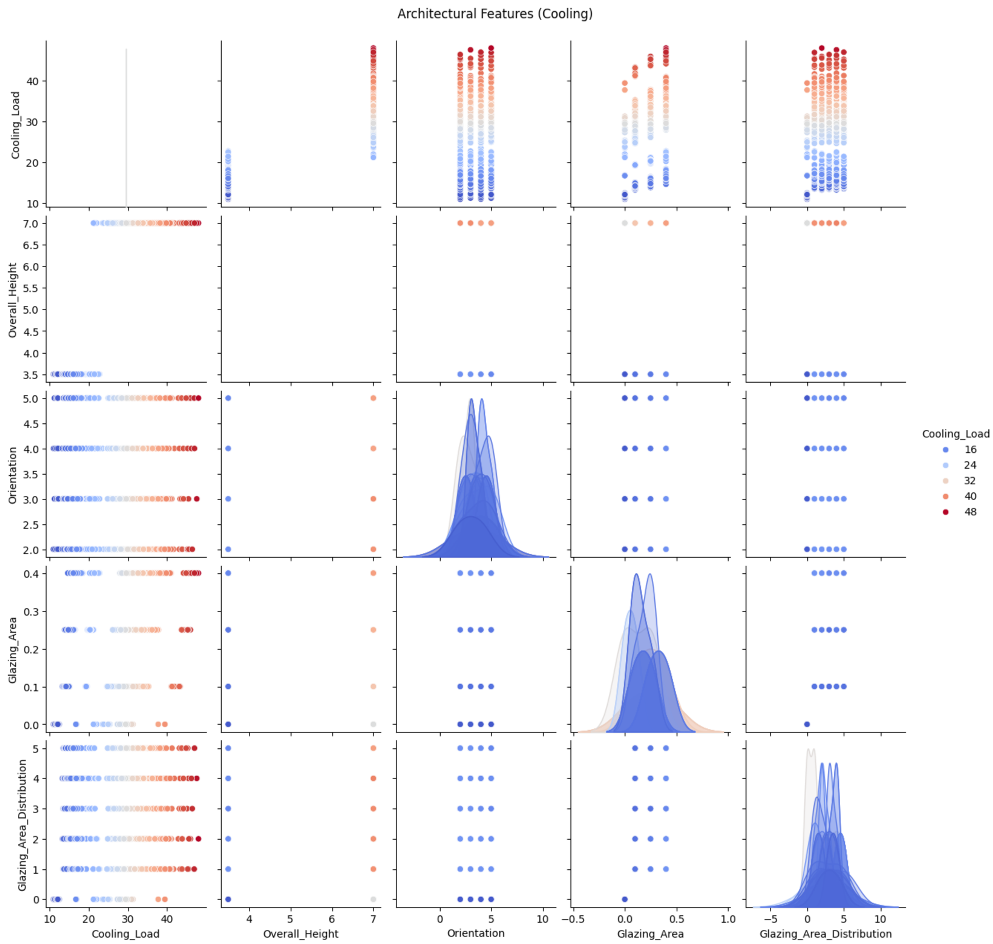

Figure 6: Scatter Plot for Architectural Features (Cooling)

Similar findings described in the Heating sections were noted. Orientation again seems to have no meaningful correlation with our other dependent variable (Cooling) and Height has some positive correlation as well as the Glazing Area. Therefore, we can proceed and drop surface area from our data since will make our model redundant and avoid collinearity.

3.4 Boxplot Findings

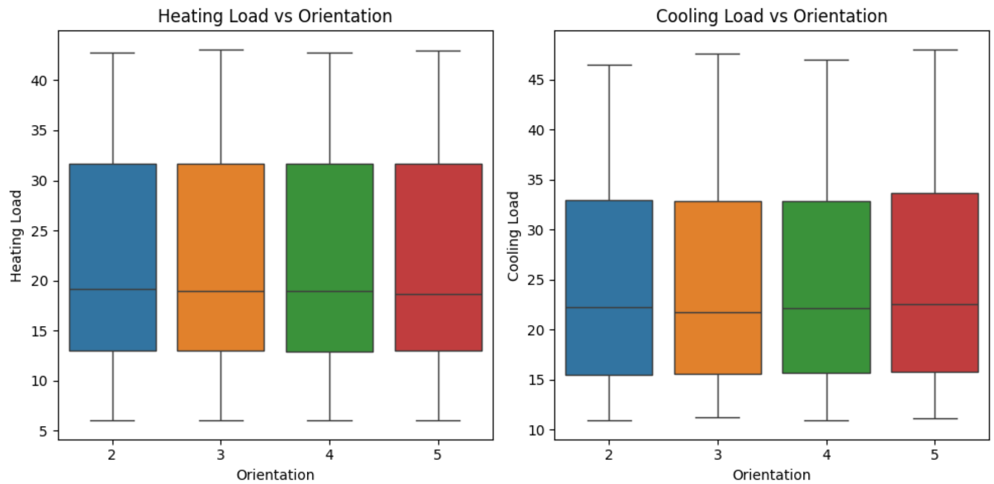

Figure 7: Boxplot for Orientation

By taking a look at both our boxplots for Heating Load vs Orientation and Cooling Load vs Orientation, we can see for all orientations, the boxplots look almost identical to each other, specially when we take a look at the medians. With these findings we drop this feature to reduce dimensionality in our model since it will not affect the overall result for prediction.

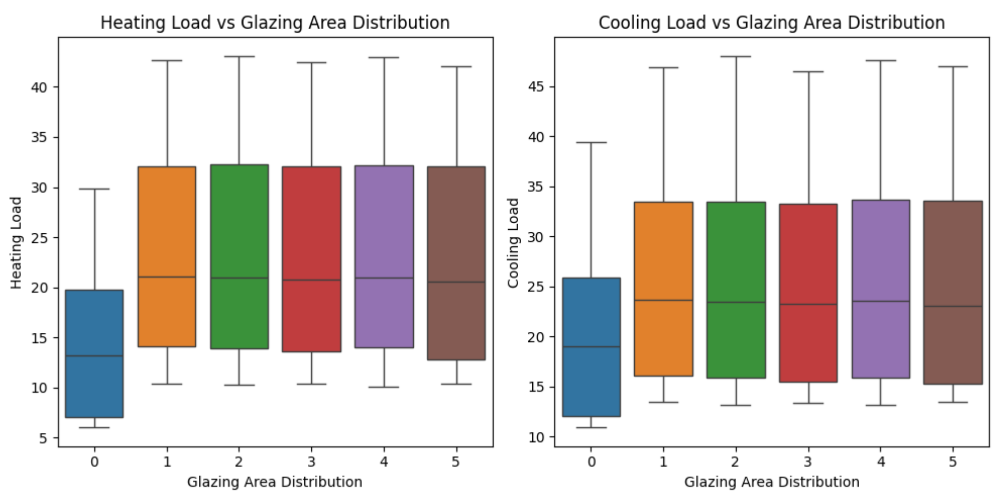

Figure 8: Boxplot for Glazing Area Distribution

For the Glazing Area Distribution only type (0) has an impact for heating and cooling which means that no windows placed nowhere makes it the construction more cold but this is not ideal for real world application. This time we are going to leave this variable to not list this information, the rest of the boxplots are identical to each other.

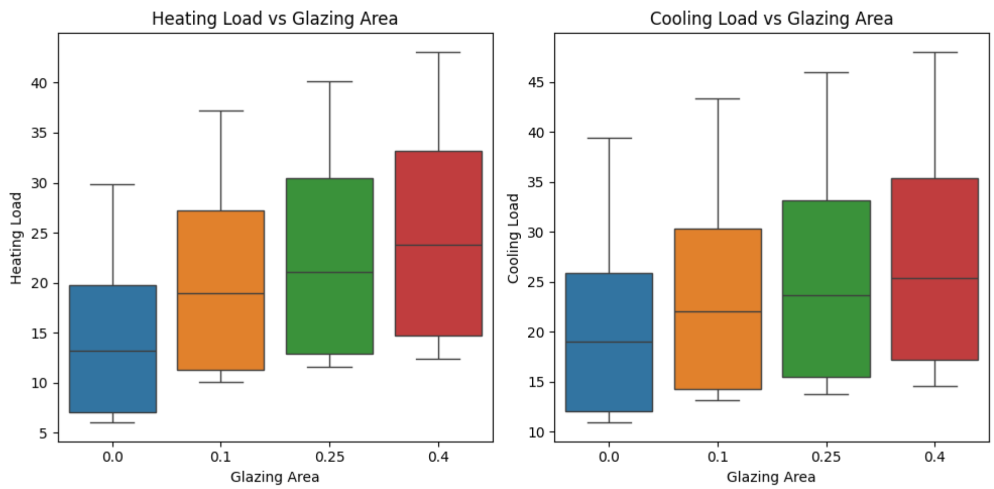

Figure 9: Boxplot for Glazing Area

For the Glazing Area we can see the area has a strong influence in our model where the less area percentage the cooler it becomes and more glazing area increase the heating and cooling loads.

3.5 EDA Summary

After performing EDA and dive deeper into our data we can see that geometric variables dominate energy performance for each dependent variable. These findings suggest that Relative Compactness has a strong inverse relationships with both loads, while Glazing Area and Height increase energy demand. The rest of the variables show mostly a non-linear trend which can be later assess when we do regression analysis.

4 Regression Modeling

4.1 Linear Regression

First, we dive deep into Linear Regression which can give us better insight on how our model behaves as well as making it mathematical simple to later apply more complex models.

4.1.1 Heating Load

The model for our Linear Regression Analysis is defined as the following:

$$Y_1 = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + \beta_5 x_{5i} + \beta_6 x_{6i} + \varepsilon_i$$

$$HeatingLoad \sim Compactness + WallArea + RoofArea + Height + GlazingArea + GlazingDistribution$$

Key Findings:

Model Fit Statistics				
$\mathbf{R^2}$ = 0.916 — **Adjusted** $\mathbf{R^2}$ = 0.916 — **F-statistic** = 1387.0 (p = 0.000) — **n** = 768				
Variable	**Coefficient**	**t-value**	**p-value**	**95% CI**
Intercept	83.93	4.41	0.000	(46.60, 121.27)
Relative Compactness	-64.77	-6.30	0.000	(-84.96, -44.59)
Wall Area	-0.03	-2.07	0.038	(-0.05, -0.001)
Roof Area	-0.17	-5.12	0.000	(-0.24, -0.11)
Overall Height	4.17	12.35	0.000	(3.51, 4.83)
Glazing Area	19.93	24.50	0.000	(18.34, 21.53)
Glazing Area Distribution	0.20	2.92	0.004	(0.07, 0.34)
Residual Diagnostics: Omnibus p = 0.000, Jarque-Bera p = 0.000, Durbin-Watson = 0.654				

Table 2: Complete Regression Results for Heating Load

- **Model Performance**: Excellent explanatory power with 91.6% of variance explained and strong overall significance (F = 1387.0, p = 0.000)
- **Variable Significance**: All predictors are statistically significant at the 5% level:
  - **Glazing Area**: Strongest positive effect ($\beta = 19.93$, p = 0.000)
  - **Relative Compactness**: Strong negative relationship ($\beta = -64.77$, p = 0.000)
  - **Overall Height**: Significant positive impact ($\beta = 4.17$, p = 0.000)
  - **Roof Area**: Moderate negative effect ($\beta = -0.17$, p = 0.000)
  - **Wall Area**: Small negative influence ($\beta = -0.03$, p = 0.038)
  - **Glazing Distribution**: Modest positive effect ($\beta = 0.20$, p = 0.004)
- **Practical Implications**:
  - Building compactness is the dominant factor in reducing heating demand
  - Glazing characteristics play a crucial dual role in energy requirements
  - Vertical dimension and surface areas meaningfully contribute to heating load
- **Model Limitations**:
  - **Non-normal Residuals**: Omnibus (p = 0.000) and Jarque-Bera (p = 0.000) tests indicate significant departure from normality
  - **Autocorrelation**: Durbin-Watson statistic of 0.654 shows strong positive autocorrelation (ideal range: 1.5–2.5)
  - **Statistical Concerns**: These violations may affect reliability of p-values and confidence intervals

Conclusion: Despite excellent explanatory power and statistically significant predictors, the model exhibits violations of key OLS assumptions. While it provides valuable insights for building design optimization—particularly regarding compactness and glazing management—results should be interpreted with appropriate statistical caution. Future analysis should consider robust standard errors or alternative modeling approaches to address these diagnostic issues.

4.1.2 Cooling Load

The model for our Linear Regression Analysis is defined as the following:

$$Y_2 = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i} + \beta_5 x_{5i} + \beta_6 x_{6i} + \varepsilon_i$$

$$CoolingLoad \sim Compactness + WallArea + RoofArea + Height + GlazingArea + GlazingDistribution$$

Model Fit Statistics				
$\mathbf{R^2} = 0.888$ — **Adjusted** $\mathbf{R^2} = 0.887$ — **F-statistic** = 1002.0 (p = 0.000) — **n** = 768				
Variable	**Coefficient**	**t-value**	**p-value**	**95% CI**
Intercept	97.67	4.70	0.000	(56.90, 138.44)
Relative Compactness	-70.79	-6.31	0.000	(-92.83, -48.75)
Wall Area	-0.04	-3.13	0.002	(-0.07, -0.02)
Roof Area	-0.18	-4.74	0.000	(-0.25, -0.10)
Overall Height	4.28	11.62	0.000	(3.56, 5.01)
Glazing Area	14.72	16.57	0.000	(12.97, 16.46)
Glazing Area Distribution	0.04	0.53	0.594	(-0.11, 0.19)
Residual Diagnostics: Omnibus p = 0.000, Jarque-Bera p = 0.000, Durbin-Watson = 1.096				

Table 3: Complete Regression Results for Cooling Load

The cooling load model demonstrates strong performance with $R^2 = 0.888$, indicating that 88.8% of the variance in cooling load is explained by the predictors. The highly significant F-statistic (1002.0, p = 0.000) confirms the overall model significance.

Key Findings for Cooling Load:

- **Model Performance**: Strong explanatory power with 88.8% of variance explained, though slightly lower than the heating load model (91.6%)
- **Variable Significance**:
  - **Glazing Area**: Strong positive effect ($\beta = 14.72$, p = 0.000), though less impactful than for heating load
  - **Relative Compactness**: Strong negative relationship ($\beta = -70.79$, p = 0.000), similar magnitude to heating model
  - **Overall Height**: Significant positive impact ($\beta = 4.28$, p = 0.000)

- **Roof Area**: Moderate negative effect ($\beta = -0.18$, p = 0.000)
- **Wall Area**: Small negative influence ($\beta = -0.04$, p = 0.002)
- **Glazing Distribution**: Not statistically significant ($\beta = 0.04$, p = 0.594)

- **Comparative Insights**:
  - Building compactness strongly reduces both heating and cooling demands
  - Glazing area affects cooling load less than heating load (14.72 vs 19.93)
  - Glazing distribution is insignificant for cooling but significant for heating
  - Height has similar positive effects on both heating and cooling loads

- **Model Limitations**:
  - **Non-normal Residuals**: Severe departure from normality (Omnibus p = 0.000, Jarque-Bera p = 0.000)
  - **Autocorrelation**: Durbin-Watson of 1.096 indicates positive autocorrelation, though less severe than heating model
  - **One Insignificant Predictor**: Glazing Area Distribution fails to reach statistical significance

Conclusion: The cooling load model shows strong predictive capability with most variables maintaining statistical significance. The key differences from the heating load model include reduced impact of glazing area and the insignificance of glazing distribution. Both models share similar diagnostic concerns regarding residual normality and autocorrelation, though the cooling model shows slightly better autocorrelation metrics.

4.2 Residual Analysis and Diagnostic: QQ Plots, Residuals, and Influence

These diagnostic plots help us visually understand our models and the violations for linear regression.

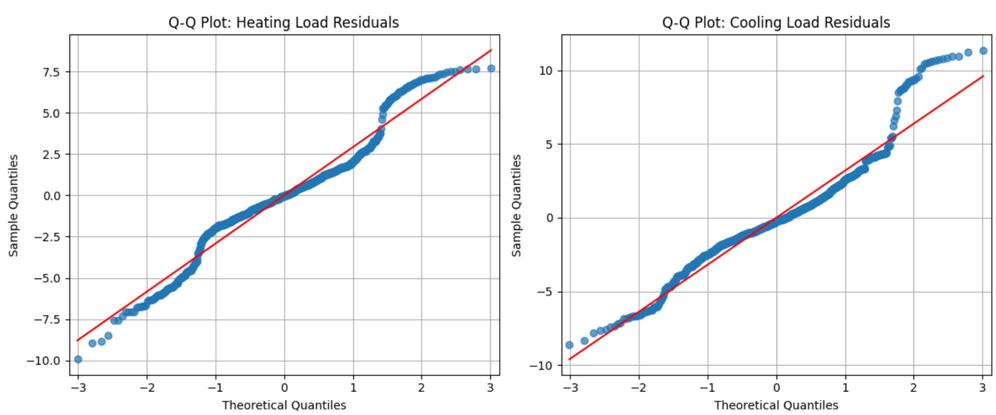

Figure 10: Q-Q Plots for Residuals

When we take a look at our Q-Q plots for our target variables we can see that both are not normally distributed. This confirms the violation of the key assumption of linear regression.

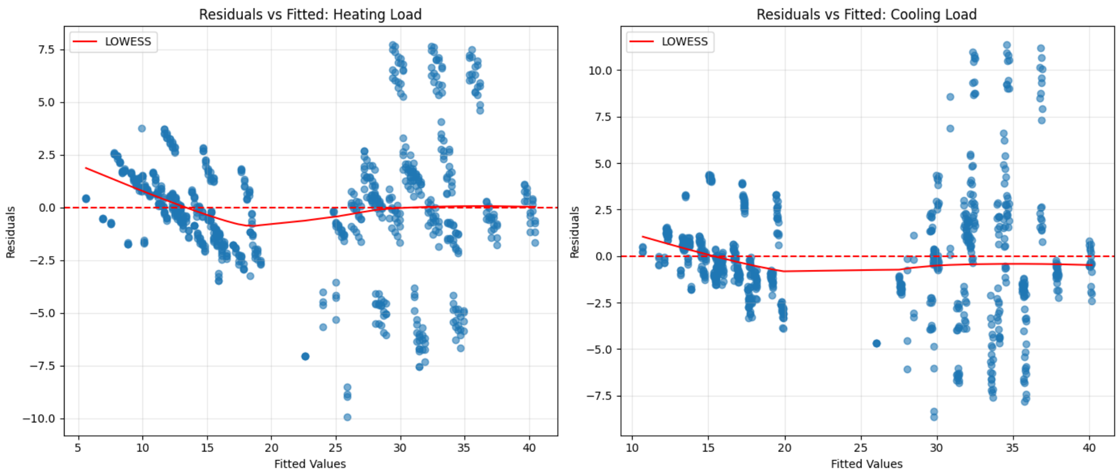

Figure 11: Residual vs Fitted with LOWESS

The residuals for this plot increases as the fitted value increase. The models become less accurate with higher energy loads.

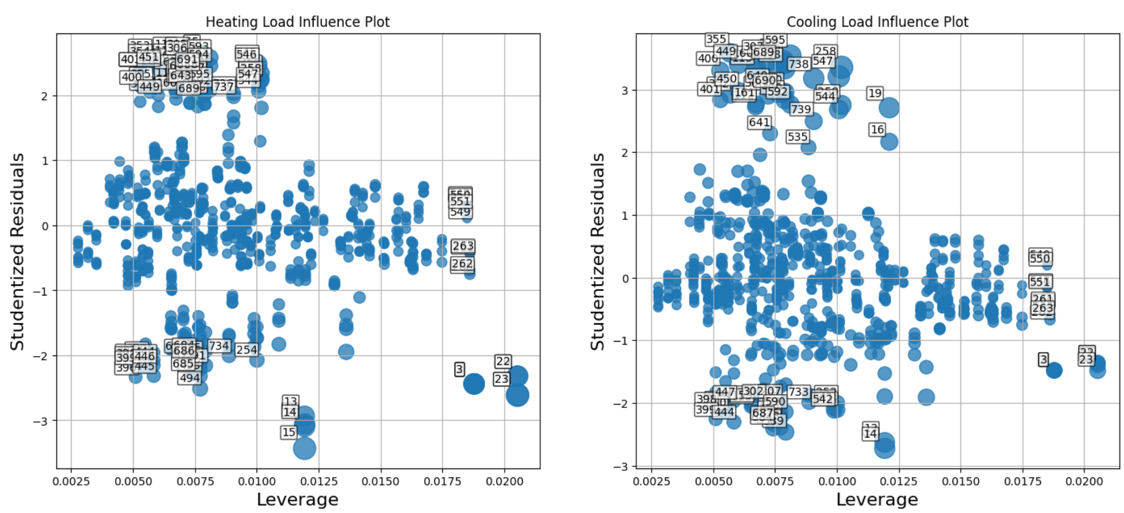

Figure 12: Influence plots showing leverage and outliers using Cook's distance for heating and cooling load. Points are sized by Cook's distance, highlighting influential observations.

When we take a look at cooks distance and influence we can see a small number of highly influential observations, specially for the Cooling Load some observations go over to standard deviations.

4.3 Random Forest

Random Forest might not be one of the simple models but since our Linear Regression model violates some assumptions, Random Forest can capture non linear pattern and capture complex interaction between our features. All Random Forest model has 100 estimators

4.3.1 Heating Loading

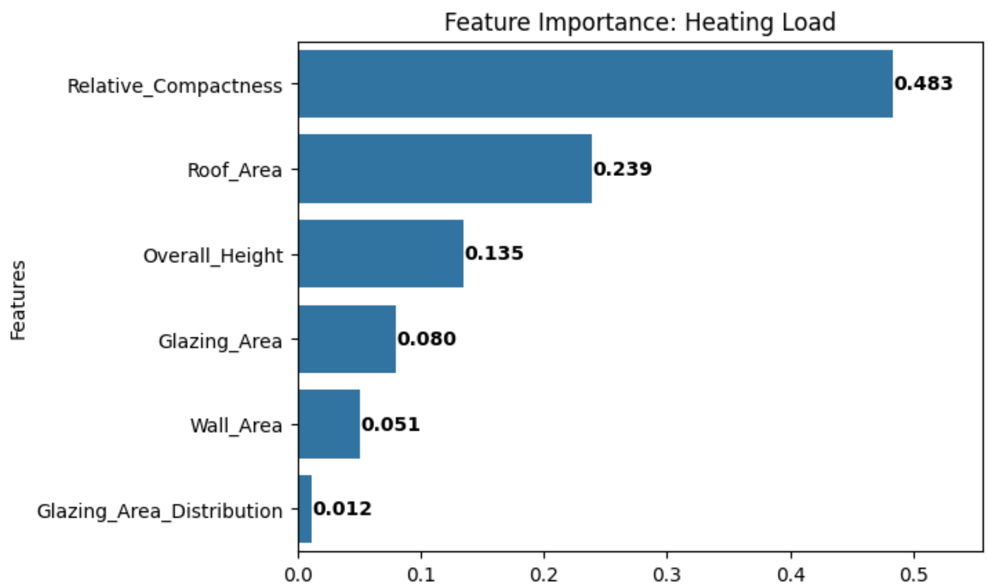

Figure 13: Random Forest Feature Importance

Random Forest feature importance rankings provide robust interpretability:

1. **Relative Compactness** (highest impact)
2. **Roof Area**
3. Overall Height
4. Glazing Area
5. Wall Area
6. Glazing Area Distribution (negligible)

This ranking aligns with thermodynamic expectations seem previously where shape dominates energy performance. Compactness has the highest effect for heating whereas Glazing Area Distribution indicates has almost no impact.

4.3.2 Cooling Loading

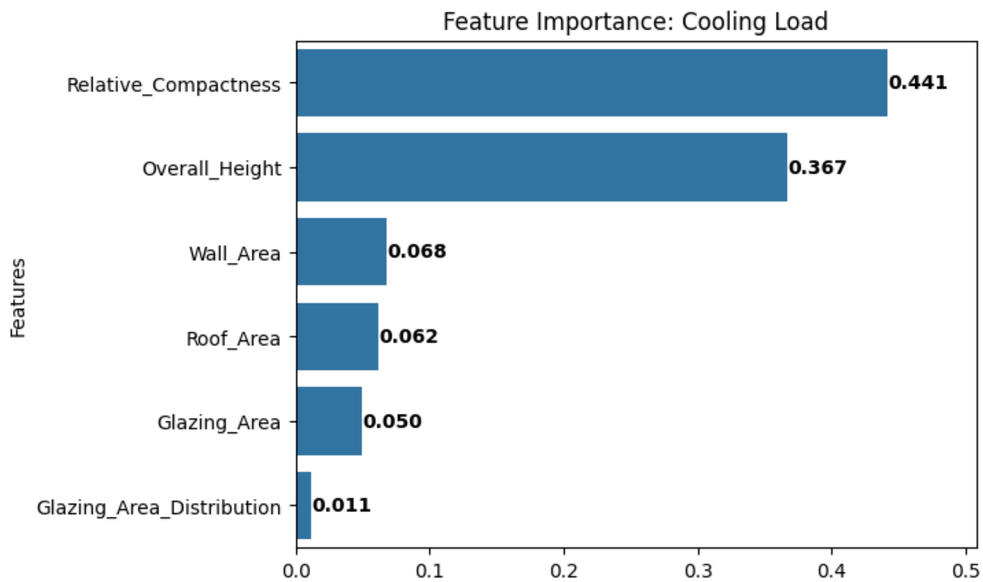

Figure 14: Random Forest Feature Importance

Random Forest feature importance rankings provide robust interpretability:

1. **Relative Compactness** (highest impact)
2. **Overall Height**
3. Wall Area
4. Roof Area
5. Glazing Area
6. Glazing Area Distribution (negligible)

For Cooling Loads it follows similar patterns as the previous where Compactness has a strong influence for cooling whereas the Glazing Area Distribution has not big impact which is negligible.

4.4 Linear Regression vs Random Forest

Linear Regression provides interpretable coefficients but fails to capture non-linear patterns, on the other hand Random Forest can capture these nonlinear interactions. Next we are going to compared them side by side.

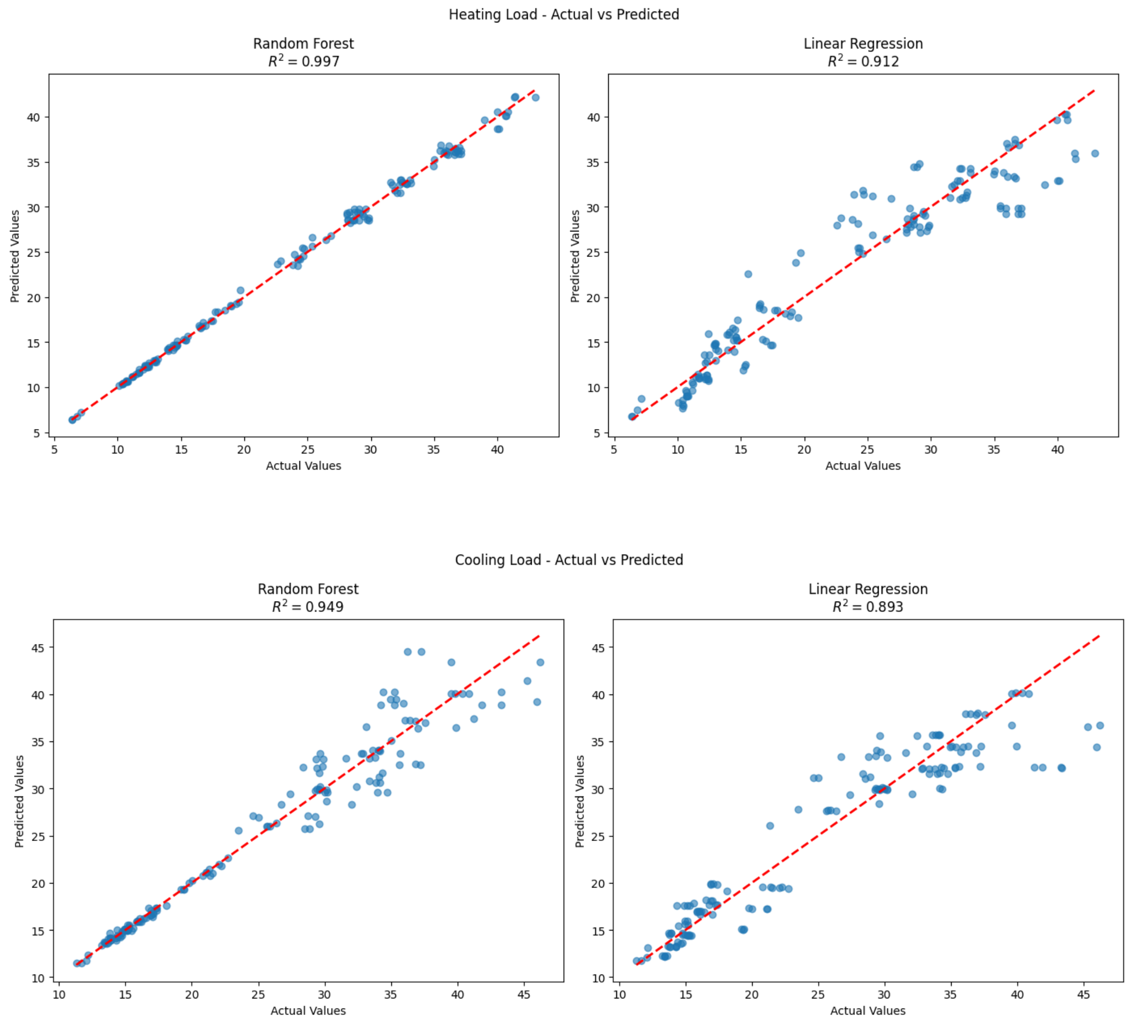

Figure 15: Heating and Cooling Load Random Forest vs Linear Regression Performance Comparison with R^2

We can see that Random Forest does a good job predicting our data where Heating Load has a better fit compared to Cooling but still better that Linear Regression. The data points are very tight around the predicted line.

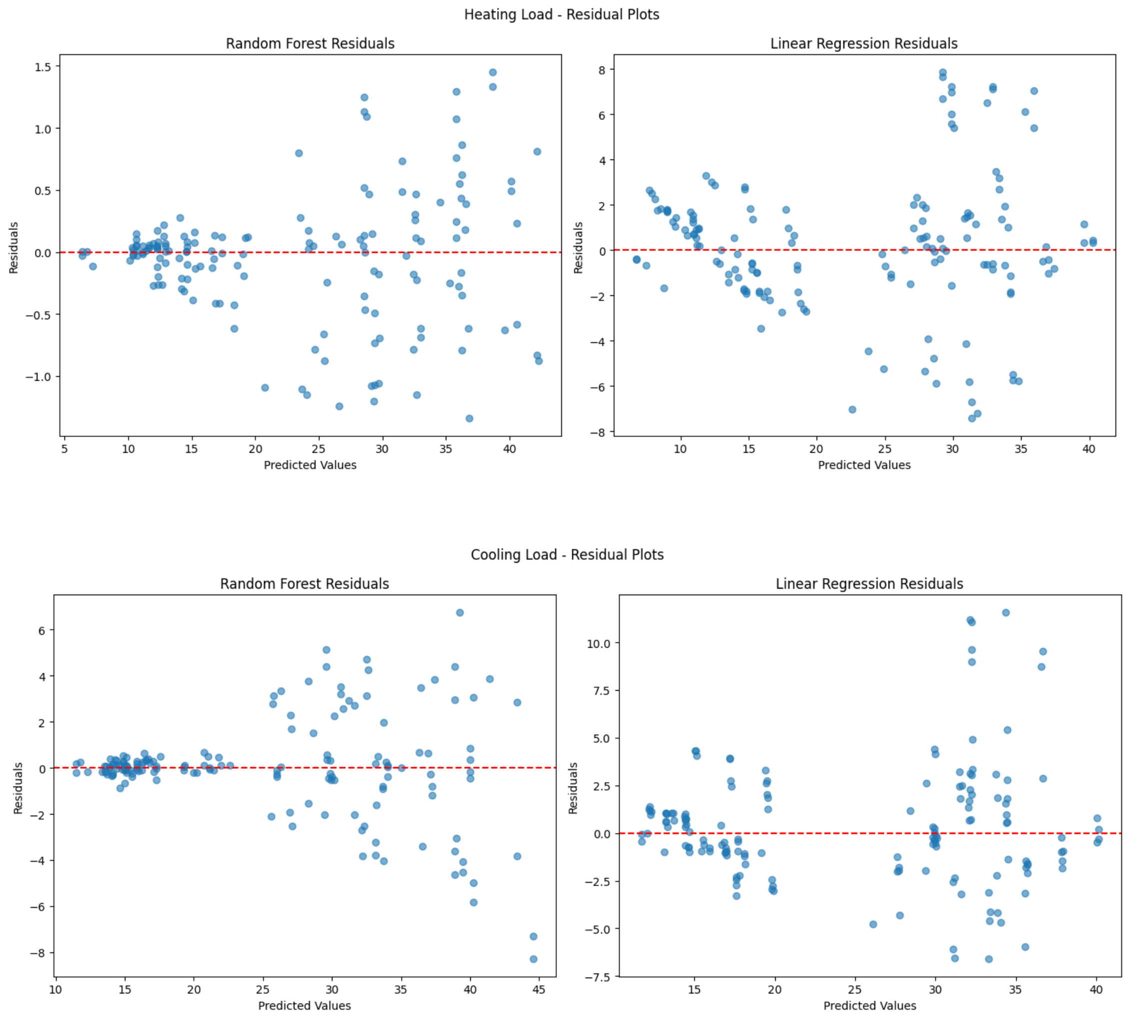

Figure 16: Heating and Cooling Load Random Forest vs Linear Regression Residuals

Our Figure 16 reveals how in Random Forest the residuals remain more tightly around zero making more stable predictions whereas Linear Regression fan outward and even more at high values which demonstrates underestimation for nonlinear features.

5 Linear Regression, Random Forest, Lasso, and Ridge Comparison

This model comparison shows how well different models such as Linear Regression, Tree based (Random Forest), Regularization based models (Lasso and Ridge) to reduce noise performs with our data. The performance of our models can captured using the R^2, MAE and RMSE.

- $\mathbf{R^2}$ measures how much variation is explained in our model. (Higher better)
- **MAE** help us measure the average size of the prediction error (Lower better)
- **RMSE** also measures error but is more extract for these mistakes.

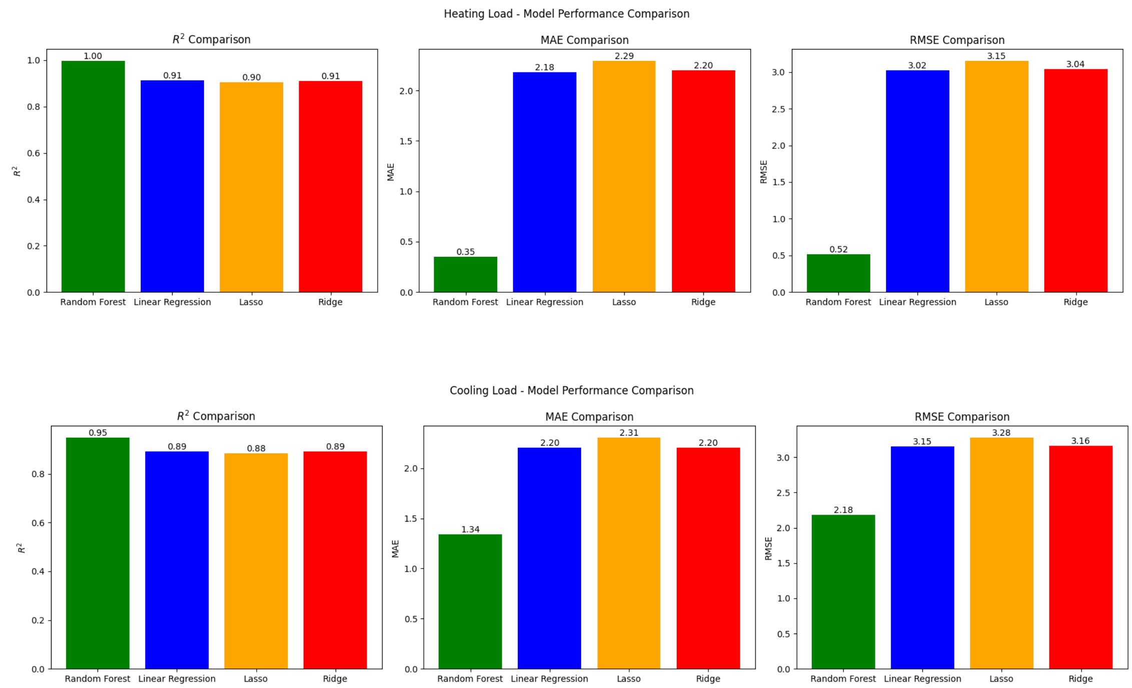

Figure 17: Heating and Cooling Load Model Performance Comparison with R^2, MAE, and RMSE

When we compared all metrics we can see that Random Forest is the best model for both our target variables having almost a perfect one rounded R^2 compared to the other models. Even though that the other models such as Linear Regression also has a high R^2 previously shown plots demonstrated that it violates core assumptions. Despite this, linear models provides helpful insight but the results should be seen as a descriptive trend not as a strong statistically inference.
Another observation that we can see is that the prediction errors for Random Forest and very low compared to the others. We can see that regularization does slightly increase the performance but it does not solve problems for non-linear data.

6 Interpretation and Engineering Insights

6.1 Compactness

After several test, Compactness consistently emerge as the strongest factor affecting energy performance. Therefore, buildings with more compact geometry have less external surface relative to their internal volume reducing the amount of heat they lose in the winter and the the heat adsorbed in the summer.

6.2 Building Height

The building height also has strong impact where as the height increases it leads to greater conditioning requirements. Even when two buildings share the same floor area the taller one will demand more heating and cooling.

6.3 Glazing

Glazing introduces predictible but nonlinear patterns. Small window areas have limited thermal effect and daylight benefits but when the percentage increase over 25% of the floor area both heating and cooling loads increase substantially. Therefore, even though its not the most stronger variable it still relevant for decision making.

6.4 Orientation

For symmetrical designs and neutral climates, orientation has negligible effect. This contradicts the common assumption that orientation is decisive. This could be due to simulated environments where the solar and neutral wind exposure are symmetrical.

7 Conclusion

Tree-based ensemble model outperformed all the other approaches for both target variables. Unlike linear models, Random Forest can capture nonlinear interactions reflecting real thermodynamics. Linear models can help us learn more about our model making it more interpretable.
Based on this analysis we can confidently say that decision can be made on this hierarchy of influence

1. Geometry
2. Envelope Exposure (Wall, Roof)
3. Glazing
4. Orientation

Based on the data suggestions we can say:

- Favor compact forms during conceptual design.
- Limit glazing proportion to moderate $\leq 25\%$ unless supported with shading or insulation.
- Avoid excessive height without envelope optimization or balance it with envelope quality.
- Use machine learning tool for early-stage building energy estimation before relying on a HVAC systems.

This analysis demonstrates how combining regression analysis can help us guide and learn about energy design strategies.